

ME 300 Thermodynamics – Summer 2021

Exam 1

Part I) _____ (30 pts)

(Part II) _____ (40 pts)

(Part III) _____ (30 pts)

TOTAL _____

SOLUTIONS

34 min

Directions

1. Open the online answers portion of the exam and download the exam (this document).
2. Make sure to show all your work!
 - a. *If you are using a tablet*, solve the problems directly on this document, showing all your work and writing as clearly as possible.
 - b. *If you are writing on paper*, solve the problems on a separate piece of paper (you do not have to print the exam), showing all your work and writing as clearly as possible.
3. Questions that require answers to be submitted to the online answers page on Canvas are written in *red italics* below as a way to remind you of what should be entered. All sketches and plots should be clearly drawn on your working paper and submitted with the upload.
4. If you have questions, please use the private chat function in Zoom to send a message directly to Dr. O'Connor so as not to disturb the other students. DO NOT UNMUTE AND ASK YOUR QUESTION ALOUD – this will be distracting to the other students.
5. Once the answers portion of the exam closes, scan or take very nice pictures of your work and upload it to the upload portion of the exam in a PDF document. I trust that you will not change anything on your exam after the answers portion of the exam closes. I can't enforce this – I'm really just relying on you to do the right thing. Please keep to the honor code.
6. The timeline for the exam will be the following (all times in EDT, Friday, June 25):
 - a. Exam opens (answers portion and upload portion): 11:00 AM
 - b. Answers portion of the exam closes: 12:30 PM
 - c. Upload portion of the exam closes: 12:45 PM
7. You can anonymously report any incidents of cheating using the following Google form:
<https://forms.gle/E4wFjDtvNuginVkP9>

Academic Integrity Pledge

I pledge that I am adhering to Penn State's Code of Conduct when taking this exam.

- I will not communicate with anyone during the exam except Dr. O'Connor.
- I will use only the allowed materials (calculator, notes, tables, and book), I will only access the internet to download the exam, view the e-book and notes on Canvas (no videos), and to submit my answers and upload the exam.
- I will not change any of my written answers after the exam but before I upload my work.
- I will not cheat in any way.

It is important to me to be a person of integrity and that means that all answers on this exam are my individual answers. *In the online answers, please type your full legal name to agree to the academic integrity pledge.*

Part I: Concept Questions

1. (10 points) In my lab, I have four tanks of ideal gases (see gas chart below), each with the same volume $[m^3]$ and at the same temperature $[K]$. Each tank has the same mass $[kg]$ of gas in it.

Gas	Molecular Weight
Oxygen (O_2)	31.9988 kg/kmol
Nitrogen (N_2)	28.0134 kg/kmol
Hydrogen (H_2)	2.016 kg/kmol
Methane (CH_4)	16.04 kg/kmol

Based off example
in Video M3.3

- a) (4 points) Order the tanks from the highest pressure (1) to the lowest pressure (4). *Answer this question in Canvas.*
b) (6 points) Explain your answer using an equation of state. *Type your answer in the box in Canvas.*

$PV = NR_uT$ +2
 $P = \frac{NR_uT}{V}$ $\rightarrow P = \frac{1}{m} \left(\frac{MR_uT}{V} \right)$ $\Rightarrow$ so $P \sim \frac{1}{m}$ $\rightarrow$ highest P is lowest m +2
 where $m = \frac{M}{N}$
 $N = \frac{M}{m}$ +2

1. H_2 +1
 2. CH_4 +1
 3. N_2 +1
 4. O_2 +1

2. (10 points) There are two piston-cylinder devices – one filled with 1 kg of water and one with 1 kg of air. Both undergo isothermal processes at 450 K from $0.002 m^3$ to $0.2 m^3$. Which one has a larger change in internal energy? Explain what's happening in both cylinders. *Type your answer in the box in Canvas.* Based off 2017 Practice exam problem I3

$T_1 = T_2 = 450 K$

$V_1 = 0.002 m^3$

$V_2 = 0.2 m^3$

$M = 1 kg$

Air

$\rightarrow$ Assume ideal gas

$\rightarrow U = U(T)$ for
ideal gases +2

$\Rightarrow \Delta \bar{U} = 0$ for
ideal gas +2

$|\Delta \bar{U}|_{\text{water}} > |\Delta \bar{U}|_{\text{air}}$ +2

Water

$\rightarrow @ 450 K \quad V_f = 0.0011232 \frac{m^3}{kg}$

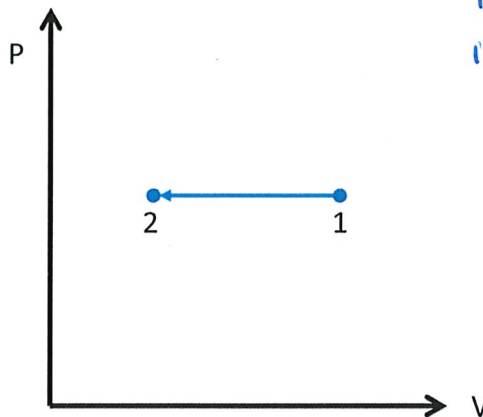
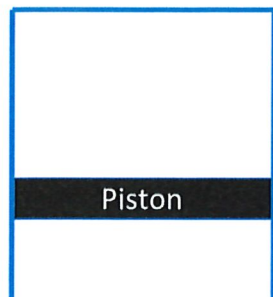
$V_g = 0.20781 \frac{m^3}{kg}$

from Table D.1 +2

$\Rightarrow$ this system is a
mixture the whole time

$\Rightarrow \Delta \bar{U} > 0$ because
 $x \uparrow$ as more vapor is
formed

3. (10 points) Analyze the P-V diagram below for the given piston-cylinder system (also pictured below), which is similar to what would be seen in an automotive engine, to answer the following four questions.



Based off example
in Video M5.3

a) (2 points) What type of process is this? *Answer the multiple-choice question in Canvas.*

- +2 ☒ A. Isothermal
☒ B. Isobaric $P = \text{const.}$
☐ C. Isochoric

b) (2 points) Which way does the piston move from State 1 to State 2? *Answer the multiple-choice question in Canvas.*

- +2 ☒ A. Up $V_2 < V_1$
☐ B. Down

c) (2 points) What is the sign of work from State 1 to State 2? *Answer the multiple-choice question in Canvas.*

- +2 ☒ A. Positive
☒ B. Negative $\Delta V < 0 \rightarrow \int_{V_1}^{V_2} P dV < 0$

d) (4 points) Explain your answer to the previous question, "What is the sign of work from State 1 to State 2?"

Type your answer in the box in Canvas.

+2 $\int_{V_1}^{V_2} P dV$

$V_2 < V_1$ and $P > 0$, so $\int_{V_1}^{V_2} P dV < 0$ +2

Part II: Short Problems

1. (20 points) A piston-cylinder device contains 2 kg of water at 420 K, where the fluid is initially a saturated liquid. Mechanical equilibrium is maintained as the piston slowly moves to State 2, where the quality of the water is 0.6. Identify any tables you use at each step of the problem. Based off EE3.

- a) (2 points) Write the assumptions necessary to solve for State 2 and the heat and work. *Enter these in the Canvas quiz – you do not need to also write them on the page.*
- b) ⁷ Calculate the change in volume in meters cubed [m^3]. *Enter this value in the Canvas quiz.*
- c) ⁵ Calculate the work in kilojoules [kJ]. *Enter this value in the Canvas quiz.*
- d) ⁶ Calculate the heat in kilojoules [kJ]. *Enter this value in the Canvas quiz.*

$$M = 2 \text{ kg}$$

$$T_1 = 420 \text{ K}$$

$$x_1 = 0$$

$$P_1 = P_2 \text{ (mech. eq.)}$$

$$x_2 = 0.6$$

a) Assume $\rightarrow$ S.C.S. ^{+1/2}

$$\rightarrow \Delta P = 0 \rightarrow \text{mechanical eq.} \quad +1/2$$

$$\rightarrow \text{quasi-steady process} \quad +1/2$$

$$\rightarrow \Delta KE = \Delta PE = 0 \quad +1/2$$

b) $\Delta V = V_2 - V_1 \rightarrow$ State 1: $x_1 = 0$ ⁺²
 $V_1 = V_f = 0.0010870 \frac{\text{m}^3}{\text{kg}}$ from D.1
 $= M(V_2 - V_1)$

$$\rightarrow \text{State 2: } x_2 = 0.6$$

$$V_2 = x_2 V_g + (1 - x_2) V_f \quad +2$$

$$= (0.6)(0.42520) + (0.4)(0.0010870)$$

$$= 0.256 \frac{\text{m}^3}{\text{kg}}$$

$$\Rightarrow \Delta V = M(V_2 - V_1) \quad +2$$
$$= (2)(0.256 - 0.0010870)$$

$\Delta V = 0.5098 \text{ m}^3$ ⁺¹

$$b) \quad {}_1\bar{W}_2 = \int_{v_1}^{v_2} P dv^{+2} \text{ where } P = P_{\text{sat}} (T = 420 \text{ K}) = 0.437309 \times 10^6 \text{ Pa}^{+1}$$

$$= MP (v_2 - v_1)^{+1}$$

$$= (2)(0.437309 \times 10^6)(0.256 - 0.0010870)$$

$${}_1\bar{W}_2 = 222951.5 \text{ J}$$

$$\boxed{{}_1\bar{W}_2 = 222.951 \text{ kJ}}^{+1}$$

$$d) \quad \Delta \bar{U} = {}_1Q_2 - {}_1\bar{W}_2^{+2}$$

$${}_1Q_2 = \Delta \bar{U} + {}_1\bar{W}_2$$

$$= M(u_2 - u_1)^{+1} + {}_1\bar{W}_2$$

From Table D.1

$$\rightarrow u_1 = u_f = 618.13 \text{ kJ/kg}^{+1}$$

$$\rightarrow u_2 = x_2 u_g + (1 - x_2)(u_f)^{+1}$$

$$= (0.6)(2556.2) + (0.4)(618.13)$$

$$= 1780.97 \frac{\text{kJ}}{\text{kg}}$$

$${}_1Q_2 = (2)(1780.97 - 618.13) + 222.951$$

$$\boxed{{}_1Q_2 = 2325.68 \text{ kJ}}^{+1}$$

2. (20 points) A rigid tank has a volume of 10 m^3 and holds 9.89 kg of water. The water is initially at 0.1 MPa , and is heated to 0.3 MPa . Identify any tables you use at each step of the problem. *Based off example in Video H5.6*

- a) ⁺² Write the assumptions necessary to solve for State 2 and the heat and work. *Enter these in the Canvas quiz – you do not need to also write them on the page.*
- b) ⁺² What is the temperature of the water at State 1 in Kelvin [K]? *Enter this value in the Canvas quiz.*
- c) ⁺² What is the temperature of the water at State 2 in Kelvin [K]? *Enter this value in the Canvas quiz.*
- d) ⁺² What is the heat added to the system from State 1 to State 2 in kilojoules [kJ]? *Enter this value in the Canvas quiz.*

Given

$$\Delta V = 0$$

$$V_1 = V_2 = 10 \text{ m}^3$$

$$M = 9.89 \text{ kg}$$

$$P_1 = 0.1 \text{ MPa}$$

$$P_2 = 0.3 \text{ MPa}$$

a) Assume $\rightarrow$ S.C.S.

$$\rightarrow \text{Rigid} \rightarrow \Delta V = 0 \quad +2$$

$$\rightarrow \text{sealed} \rightarrow \Delta M = 0$$

$$\rightarrow \text{quasi- of process}$$

$$\rightarrow \Delta KE = \Delta PE = 0$$

b) State 1 $\rightarrow P_1 = 0.1 \text{ MPa}$

$$v_1 = \frac{V_1}{M} = \frac{10}{9.89} = 1.0111 \frac{\text{m}^3}{\text{kg}} \quad +2$$

$$\rightarrow \text{Table D.2 @ } 0.1 \text{ MPa} \rightarrow v_f < v_1 < v_g \rightarrow \text{mixture} \quad +2$$

$$\Rightarrow \boxed{T_1 = T_{\text{sat}} = 372.76 \text{ K}} \quad +2$$

c) State 2 $\rightarrow P_2 = 0.3 \text{ MPa}$

$$v_2 = v_1 \rightarrow v_2 = v_1 = 1.0111 \frac{\text{m}^3}{\text{kg}} \quad +2 \Rightarrow \text{D.2} \quad v_2 > v_g @ P_2 = 0.3 \text{ MPa}$$

$$\rightarrow \text{Table D.3F} \rightarrow \boxed{T_2 = 660 \text{ K}} \quad +2 \Rightarrow \text{Vapor}$$

$$d) \Delta \bar{U} = \overset{+2}{Q_2} - \overset{+1}{\bar{W}_2}$$

$\rightarrow \Delta H = 0 \text{ so } \int P dV = 0$

$$\begin{aligned} {}_1Q_2 &= \Delta \bar{U} \\ &= M(u_2 - u_1) \end{aligned}$$

from Table D.2

$$\rightarrow \text{State 1 : } x_1 = \frac{v_1 - v_f}{v_g - v_f} = \frac{1.0111 - 0.0010432}{1.6939 - 0.0010432} = \overset{+2}{0.5966}$$

$$\begin{aligned} u_1 &= x_1 u_g + (1-x_1) u_f \quad +1 \\ &= (0.5966)(2522.6) + (1-0.5966)(417.4) \\ &= 1663.43 \frac{\text{kJ}}{\text{kg}} \end{aligned}$$

$$\rightarrow \text{State 2 : Table D.3F, } u_2 = 2944.8 \frac{\text{kJ}}{\text{kg}} \quad +1$$

$$\begin{aligned} {}_1Q_2 &= M(u_2 - u_1) \\ &= 9.89(2944.8 - 1663.43) \end{aligned}$$

$$\boxed{{}_1Q_2 = 12672.75 \text{ kJ}} \quad +1$$

Part III: Analysis (30 points)

The story: Stirling engines powered by the sun have been built for power generation in remote areas. In these systems, a working fluid is heated by a solar concentrator in the second step of the four step cycle:

- Process 1-2: Isothermal compression
- Process 2-3: Isochoric heat addition
- Process 3-4: Isothermal expansion
- Process 4-1: Isochoric heat rejection

Based off Homework 4,
Problem 1

In our Stirling engine, we're going to use air as our working fluid ($R=287 \text{ J/kg-K}$ and $c_v=714 \text{ J/kg-K}$) and assume that 10 kg of air is trapped in a piston-cylinder device, undergoing the Stirling cycle.

- a) List the assumptions you'll need to solve for the state points as well as the first law quantities (heat, work, change in internal energy). *Enter these in the Canvas quiz – you do not need to also write them on the page.*
- b) Let's start with the heat transfer step (Process 2-3). The average global solar irradiance from the sun is 1000 W/m^2 – this is the power that the sun delivers to the Earth in the form of heat. Assuming the solar collector area is 10 m^2 , how much heat does the sun deliver to the cycle if Process 2-3 takes 5 minutes? What is the temperature after this heat addition step? Hint: this question is, in part, about units – if you don't know how to solve this immediately, analyze the units. *Enter these values in the Canvas quiz.*
- c) Use the state points below to solve for the remainder of the states (P in [kPa], V in [m^3], and T in [K]). Note: if you were unable to solve part b, solve the problem symbolically for partial credit. *Enter these values in the Canvas quiz.*

State	Pressure [kPa]	Volume [m^3]	Temperature [K]
1	120005.23	7.653	320
2	1200052.3	0.7653	320
3	2775758.4	0.7653	Use your value from part b
4	277575.84	7.653	740.17

- d) What is the net work of the cycle? *Enter this value in the Canvas quiz.*

Given

$$M = 10 \text{ kg}$$

$$R = 287 \frac{\text{J}}{\text{kg-K}}$$

$$c_v = 714 \frac{\text{J}}{\text{kg-K}}$$

- a) Assume $\rightarrow \Delta M = 0$
- $\rightarrow$ air = ideal gas
- $\rightarrow c_v = \text{const.}$
- $\rightarrow$ quasi-st. processes
- $\rightarrow \Delta KE = \Delta PE = 0$

$$b) \dot{Q}'' = 1000 \frac{W}{m^2}$$

$$A = 10 m^2$$

$$t = 5 \text{ min} = 300 \text{ sec}$$

Find $\rightarrow$ heat, [J]

$$Q [J] = \dot{Q}'' \left[\frac{W}{m^2} \right] \times A [m^2] \times t [s]$$

+1 +1 +1

remembering $W \sim \frac{J}{s}$

$$Q = 1000 \times 10 \times 300 \Rightarrow \boxed{Q = 3 \text{ MJ}} \quad +2$$

$$\Rightarrow {}_2\bar{W}_3 = 0 \text{ b/c isochoric} \xrightarrow{+2} \Delta \bar{U} = {}_2Q_3 \xrightarrow{+2} \text{ideal gas: } MC_v \Delta T = {}_2Q_3 \quad +2$$

$$T_3 = \frac{{}_2Q_3}{MC_v} + T_2 = \frac{3 \times 10^6}{(10)(714)} + 320 \Rightarrow \boxed{T_3 = 740.17 K} \quad +1$$

$$c) \text{ State 1: } \begin{cases} V_1 = 7.653 m^3 \\ T_1 = T_2 = 320 K \end{cases} \quad +1$$

$$P_1 = \frac{MRT_1}{V_1} = \frac{(10)(287)(320)}{(7.653)} \Rightarrow \boxed{P_1 = 120005.23 \text{ Pa}} \quad +1$$

$$\text{State 2: } P_2 = \frac{MRT_2}{V_2} = \frac{(10)(287)(320)}{0.7653} \Rightarrow \boxed{P_2 = 1200052.3 \text{ Pa}} \quad +1$$

$$\text{State 3: } V_3 = V_2 = 0.7653 m^3 \quad +1$$

$$T_3 = 740.17 K$$

$$P_3 = \frac{MRT_3}{V_3} = \frac{(10)(287)(740.17)}{0.7653} \Rightarrow \boxed{P_3 = 2775758.4 \text{ Pa}} \quad +1$$

$$\text{State 4: } V_4 = V_1 = 7.653 m^3 \quad +1$$

$$T_4 = T_3 = 740.17 K \quad +1$$

$$P_4 = \frac{MRT_4}{V_4} = \frac{(10)(287)(740.17)}{7.653} \Rightarrow \boxed{P_4 = 277575.84 \text{ Pa}} \quad +1$$

$$d) \bar{W}_{\text{net}} = {}_1\bar{W}_2 + {}_2\bar{W}_3 + {}_3\bar{W}_4 + {}_4\bar{W}_1 \quad \text{+1}$$

$\swarrow \rightarrow 0 \quad v_2 = v_3$
 $\swarrow \rightarrow 0 \quad v_4 = v_1$

$$\rightarrow {}_1\bar{W}_2 = \int_{v_1}^{v_2} P dv \quad \text{since ideal gas} \quad P = \frac{MRT}{v}$$

$$= \int_{v_1}^{v_2} \frac{MRT}{v} dv = MRT_1 \int_{v_1}^{v_2} \frac{dv}{v} = (10)(287)(320) \ln\left(\frac{0.7653}{7.653}\right)$$

+2

$${}_1\bar{W}_2 = -2114694.15 \text{ J} \quad \text{+1}$$

$\rightarrow {}_3\bar{W}_4$ is the same as above in method

$${}_3\bar{W}_4 = MRT_3 \int_{v_3}^{v_4} \frac{dv}{v} = 10(287)(740.17) \ln\left(\frac{7.653}{0.7653}\right)$$

$${}_3\bar{W}_4 = 4891353.65 \text{ J} \quad \text{+1}$$

$$\bar{W}_{\text{net}} = 2776659.5 \text{ J} \quad \text{+1}$$

